UMC 0.11 um Logic and Mixed-Mode Al Advanced Enhancement 2.5V Diode SPICE Model Document

Version 0.5 Phase 1
2016/11/23

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/05/21	Laney Fan	1. Original release
0.3_p1	2012/08/09	Laney Fan	1. Multiply 0.9 of w and l in the subckt for shrink methodology
0.4_p1	2012/10/17	Laney Fan	1. Correct from multiply 0.9 by w and l to multiply 0.9*0.9 by area.
0.5_p1	2016/11/23	Macro Chang	1. Optimize leakage current and break down voltage

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

> 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

> G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

Table 1-2 DIODE device

DIODE (twin well)	N+ to Pwell	P+ to Nwell
Model Name	dion_25_l110ae	diop_25_l110ae

1.3 Model Usage

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9SP2
Cadence SPECTRE version 15.1.0.257 64bit
Mentor Graphics ELDO version 15.4

For Synopsys HSPICE
- Copy ModelFileName.lib into your simulation directory.
-Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "ModelFileName.lib" CaseName
, where CaseName could be TT, FF, and SS.
- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model.
Example1:(for sub-circuit format model)
X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5
Example2:(for non sub-circuit format model)
D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

For Cadence Spectre
- Copy ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption1 options scale=0.9
include "ModelFileName.lib.scs" section=CaseName
, where CaseName could be TT, FF, and SS.
- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model.
x1 (Va Vc) DeviceName area=1e-10 pj=4e-5
Example2:(for non sub-circuit format model)
d1 (Va Vc) DeviceName area=1e-10 pj=4e-5

For Mentor Graphics Eldo
- Copy ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "ModelFileName.lib.eldo" CaseName
, where CaseName could be TT, FF, and SS.
- Define device condition by 'Xxx' statement for sub-circuit format model, or 'Dxx' statement for non sub-circuit format model.
Example1:(for sub-circuit format model)

X1 Va Vc DeviceName AREA=1E-10 PJ=4E-5
Example2:(for non sub-circuit format model)
D1 Va Vc DeviceName AREA=1E-10 PJ=4E-5

The available diodes are:

a. 2.5V N+ to P-Well Diode : dion_25_l110ae
b. 2.5V P+ to N-Well Diode : diop_25_l110ae

ModelFileName: l110ae_ diode25_v051

Geometry range:

This is a scalable model, which uses actual geometry that AREA and PJ in the unit of meter^2 and meter respectively. The geometry range should follow TLR document.

Geometry range (Before 90% shrink):

2.5V DION
0.5um<=WDES<=14.3um
5um<=LDES<=25um

2.5V DIOP
0.5um<=WDES<=14.3um
5um<=LDES<=25um

Temperature range:

-50 ℃ ~ +125 ℃

2 2.5V DION Diode Model Verification

2.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for DION

	Dion diode model
Mask ID	A11007
Lot ID	D5FHR.11
Wafer Number	#11

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model is extracted by devices with these two dimensions:

Junction Area Device :

Area = 5.0e-8 m^2
Perimeter = 2.0e-3 m

Junction Edge Device :

Area = 3.6e-8 m^2
Perimeter = 3.64e-2 m

2.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 2-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
V_J(V)	0	1	0.05

Reverse Diode

	Start	Stop	Step
V_J(V)	0	-15	-0.1

Diode junction CV:

Table 2-3 Diode junction CV measurement conditions

V_{BIAS} [V]	VH: pwell & VL: n+ **From 0 to 3.63 V in 0.033 V step**
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50, 0, 25, 85, 125

2.3 2.5V DION model verification

2.3.1 2.5V DION model verification at room temperature (25 ℃)

➢ ***Diode junction CV curve at 25 ℃.***

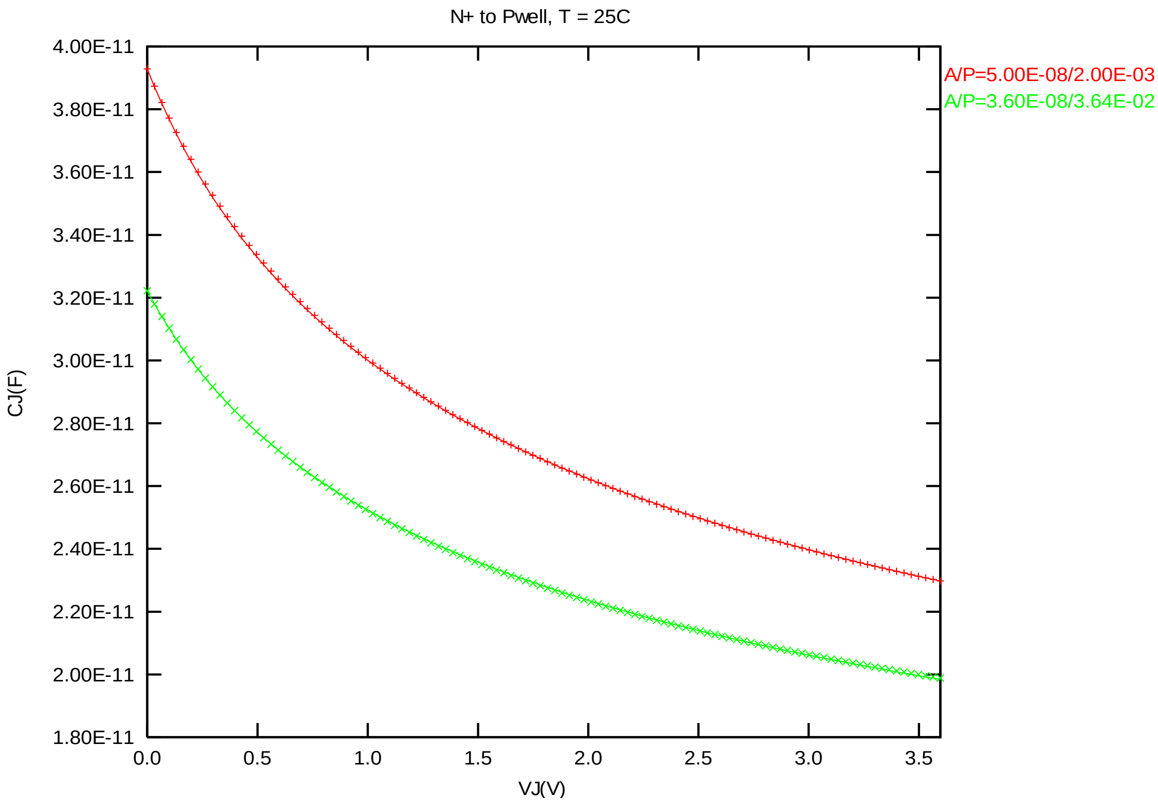
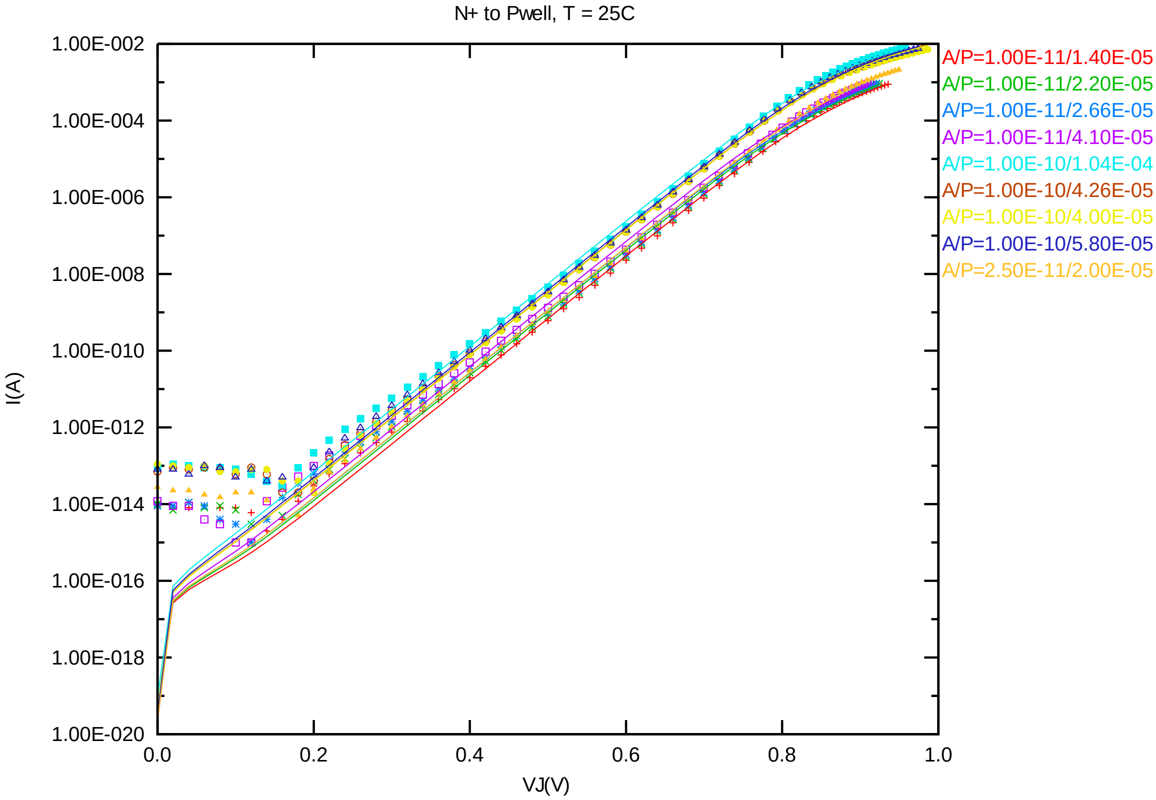


Figure 2-1 Diode junction CV curve at 25 ℃

➢ ***Forward IV curve at 25 ℃.***



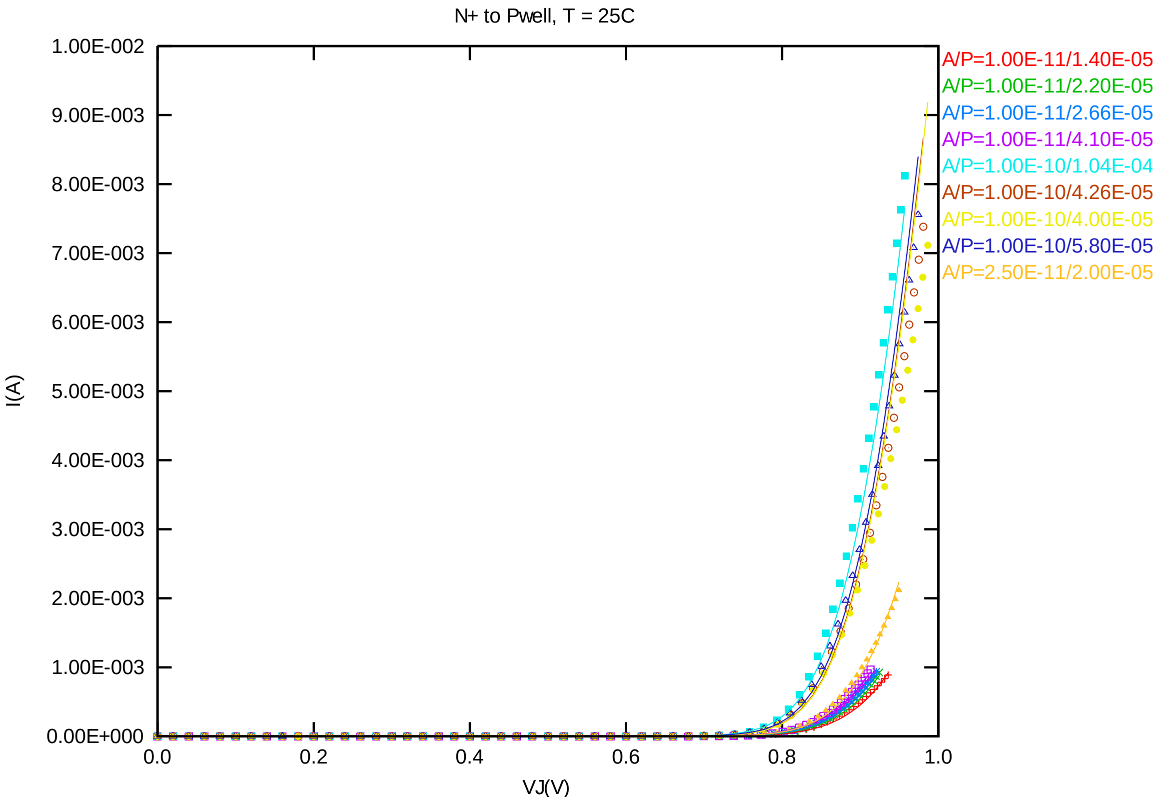


Figure 2-2 Forward IV curve at 25 ℃

- ***Reverse IV curve at 25 ℃.***

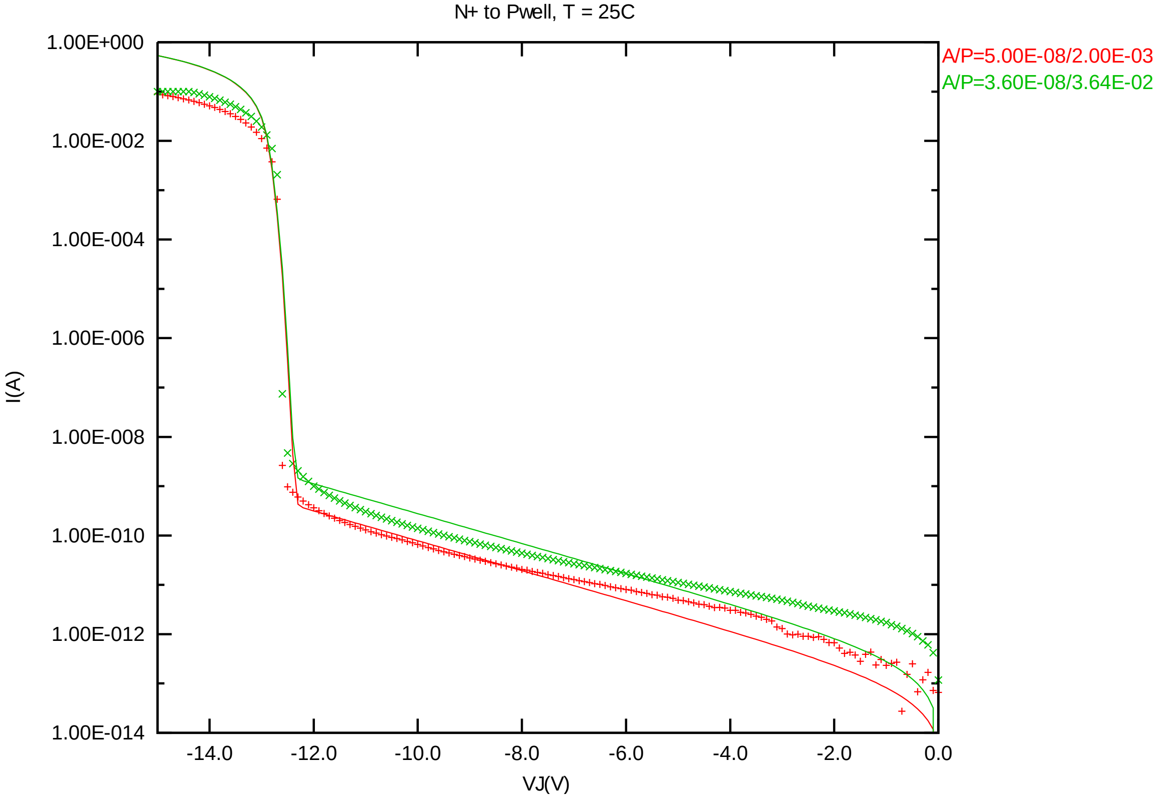


Figure 2-3 Reverse IV curve at 25 ℃

2.3.2 2.5V DION model verification at 125 ℃

➢ ***Diode junction CV curve at 125 ℃.***

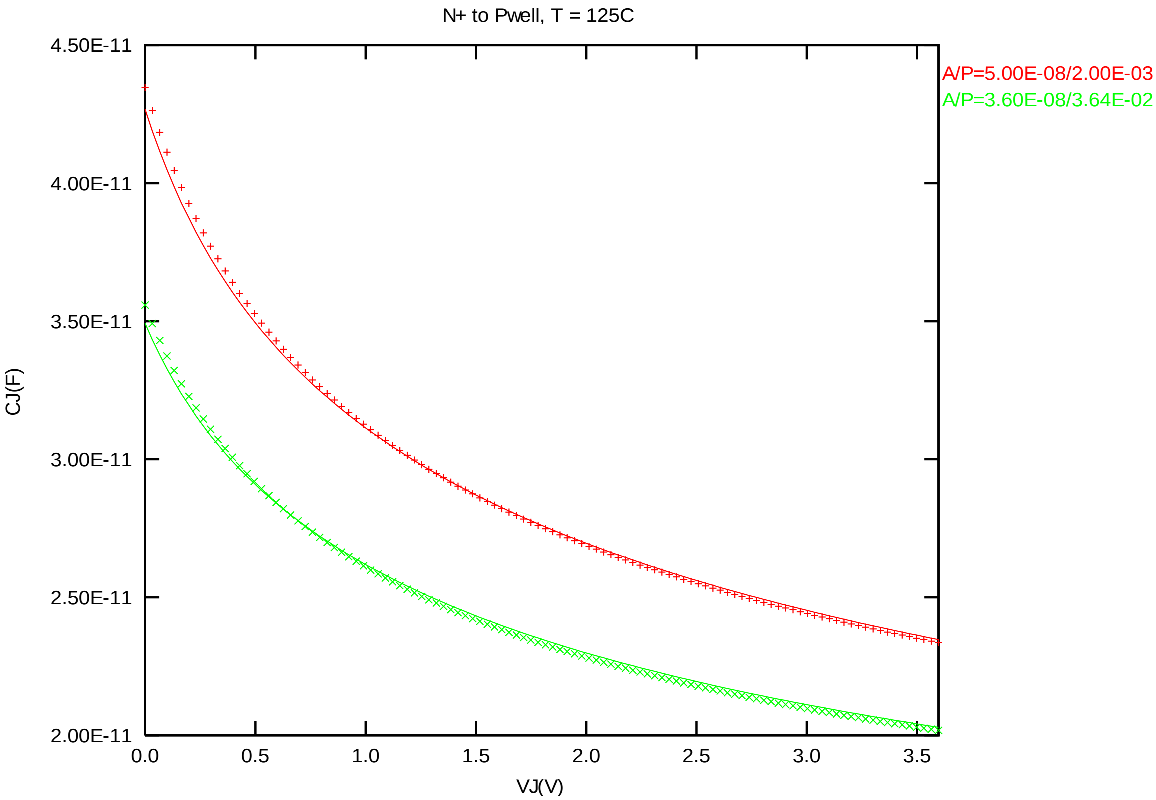
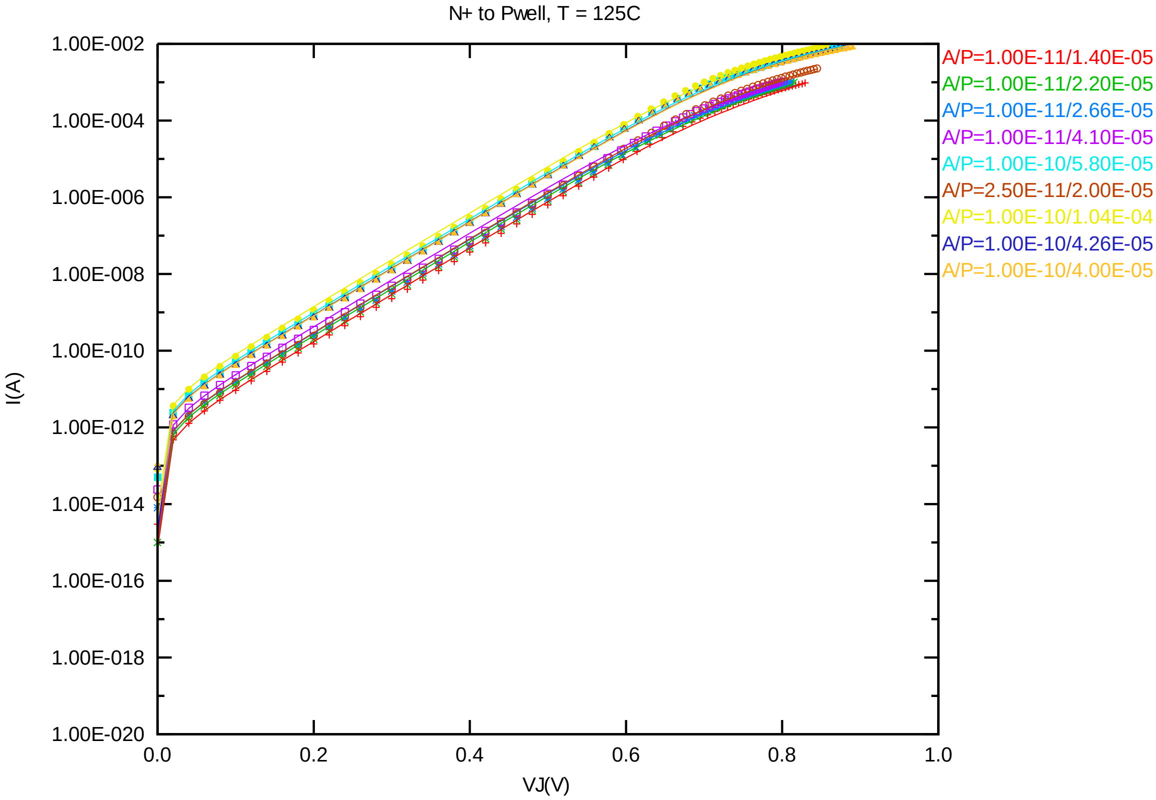


Figure 2-4 Diode junction CV curve at 125 ℃

- ***Forward IV curve at 125 ℃.***



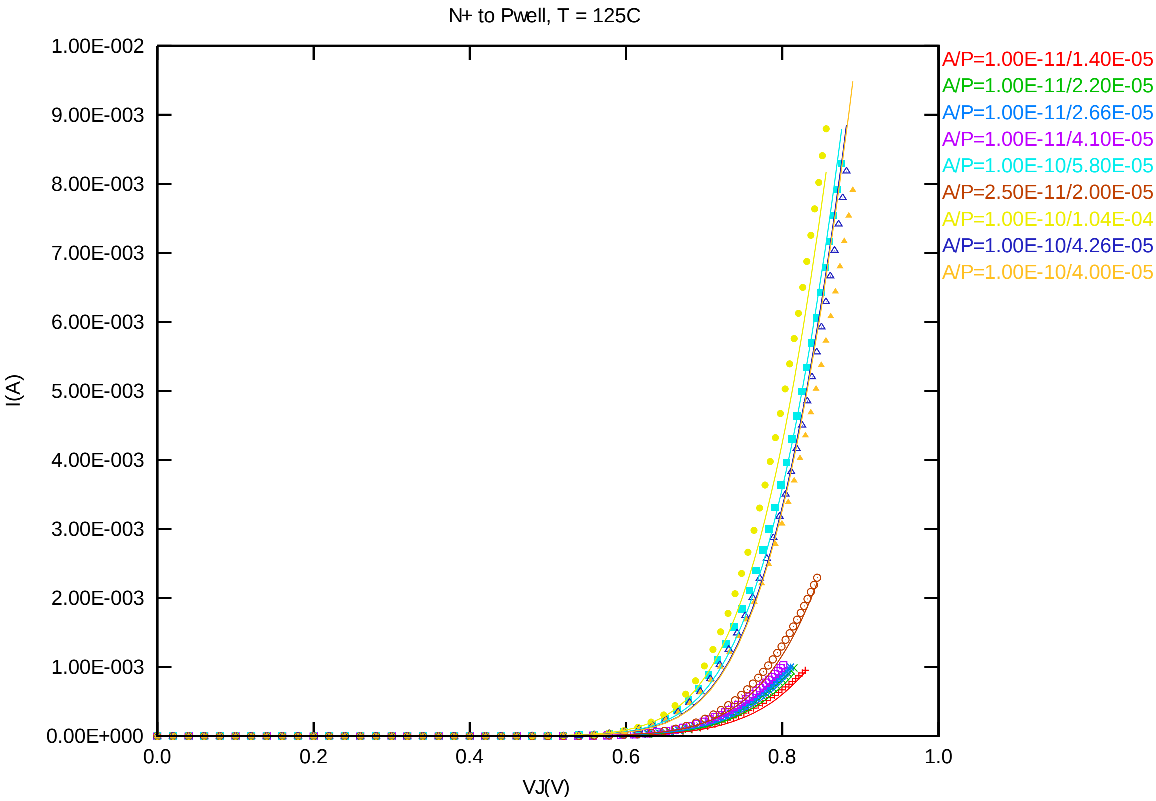


Figure 2-5 Forward IV curve at 125 ℃

- ***Reverse IV curve at 125 ℃.***

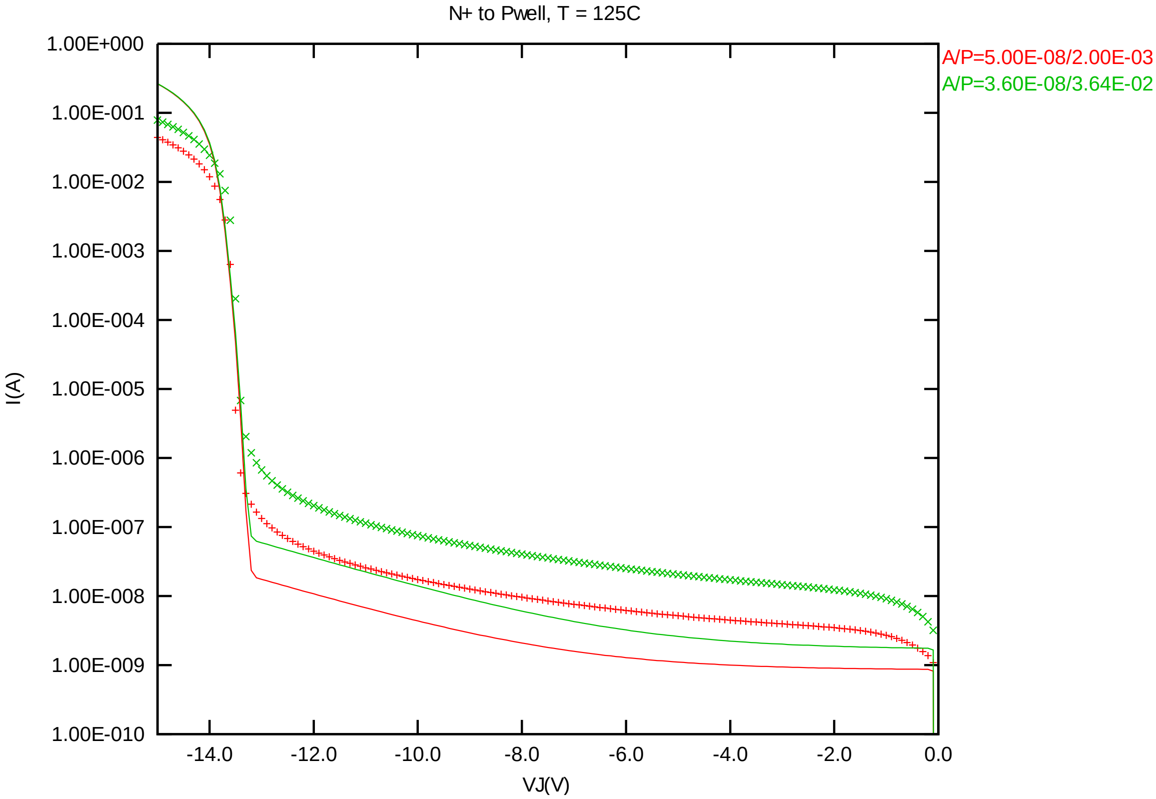


Figure 2-6 Reverse IV curve at 125 ℃

2.3.3 2.5V DION model verification at 85 ℃

➢ ***Diode junction CV curve at 85 ℃.***

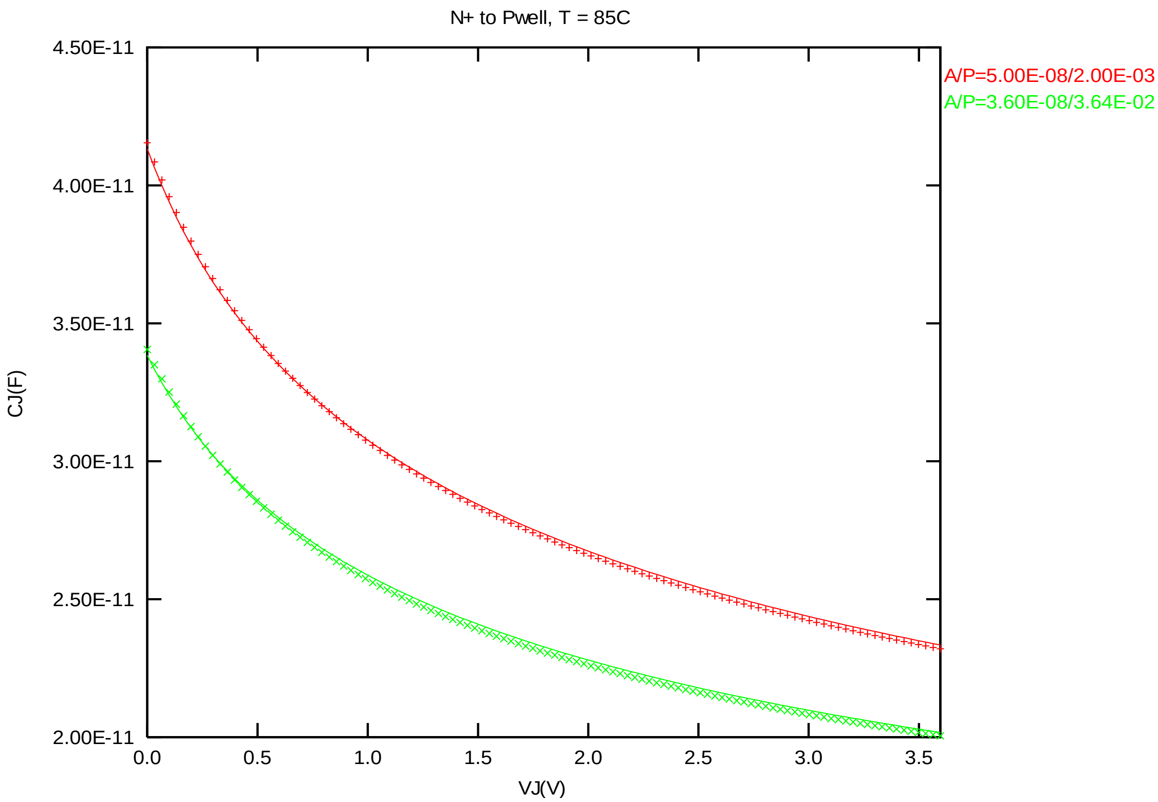


Figure 2-7 Diode junction CV curve at 85 ℃

2.3.4 2.5V DION model verification at 0 ℃

➢ ***Diode junction CV curve at 0 ℃.***

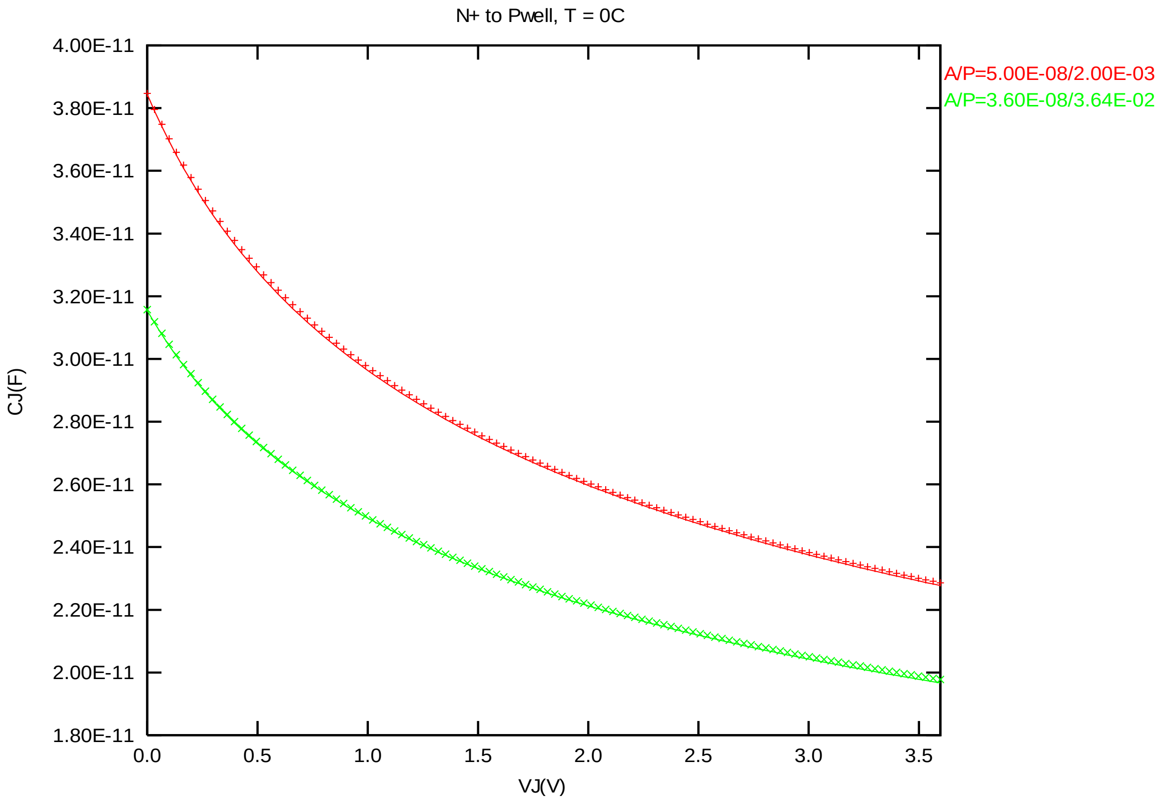


Figure 2-8 Diode junction CV curve at 0 ℃

2.3.5 2.5V DION model verification at -50 ℃

➢ ***Diode junction CV curve at -50 ℃.***

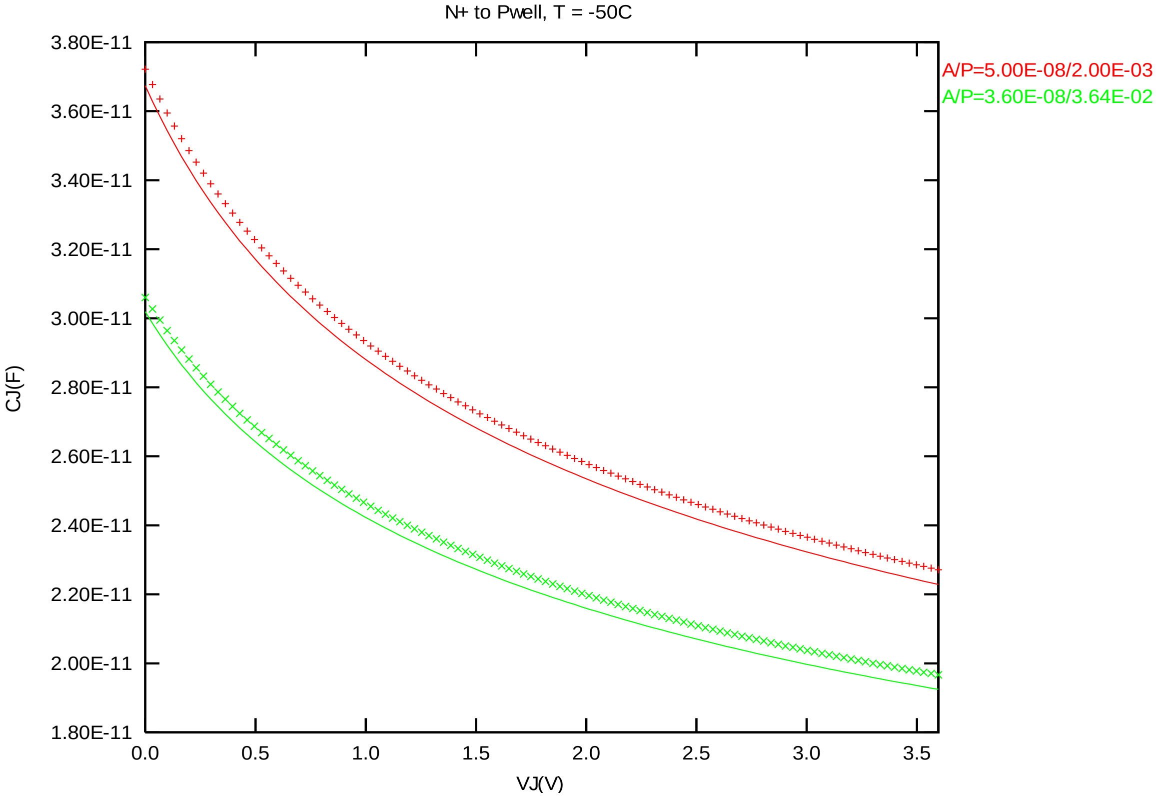
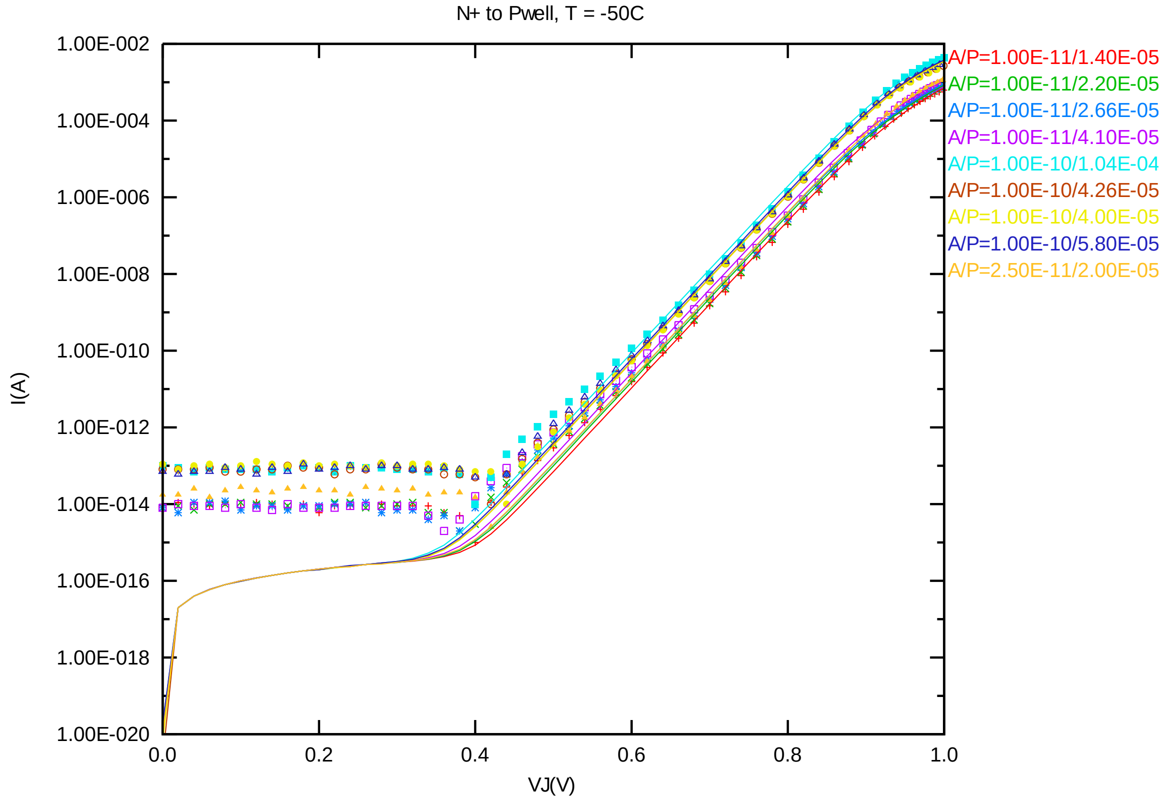


Figure 2-9 Diode junction CV curve at -50 ℃

➢ *Forward IV curve at -50 ℃.*



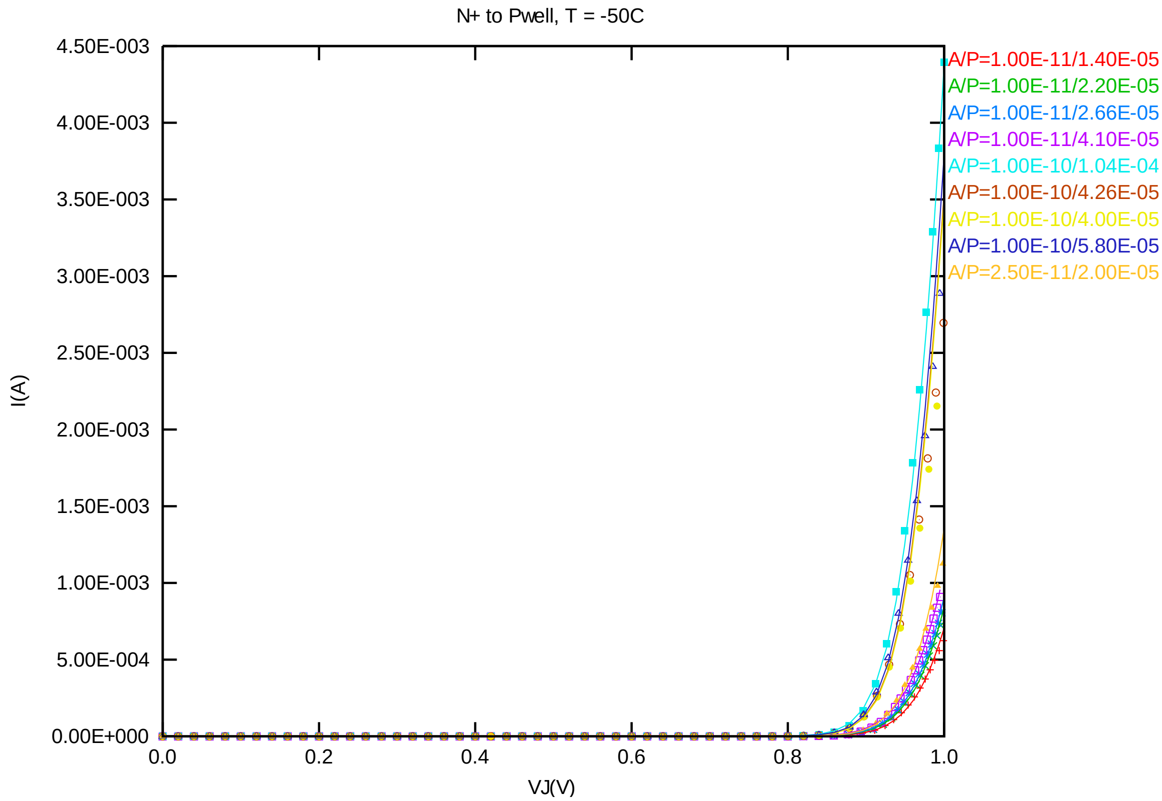


Figure 2-10 Forward IV curve at -50 ℃

- ***Reverse IV curve at -50 ℃.***

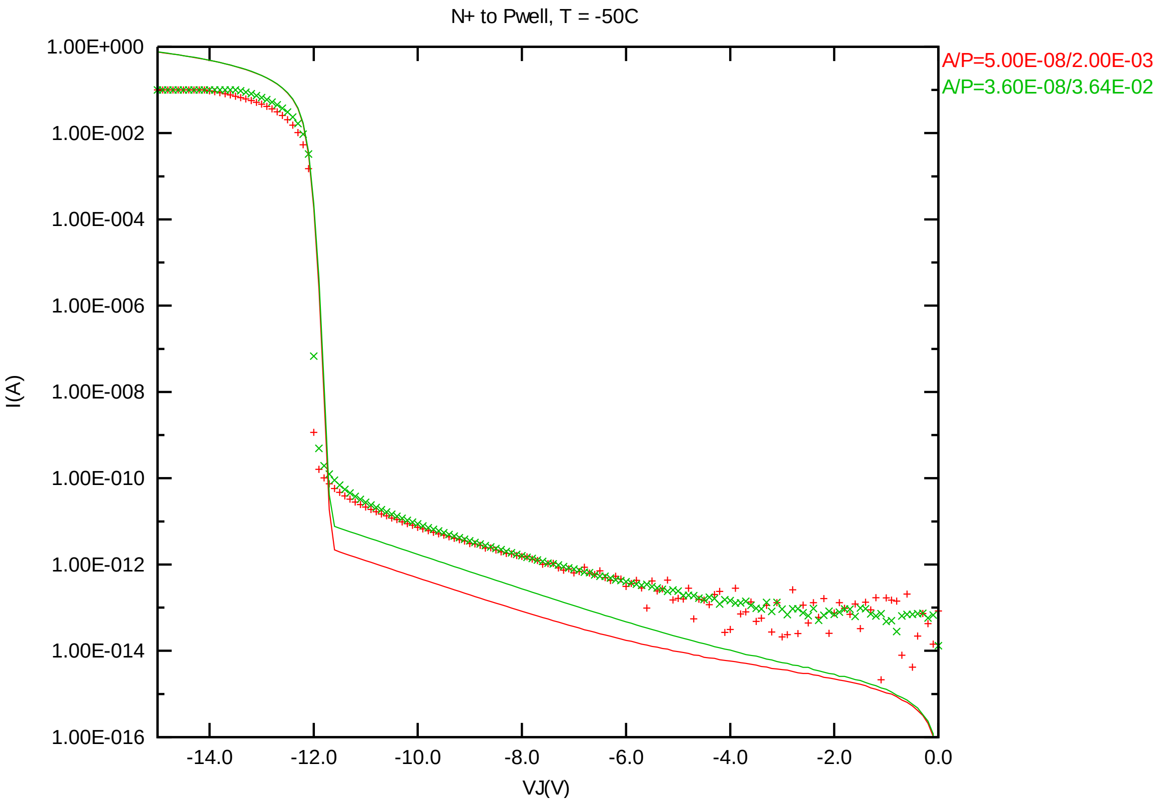


Figure 2-11 Reverse IV curve at -50 ℃

2.3.6 2.5V DION model verification for temperature effect

- ***Diode junction CV curve at 25 ℃, 125 ℃, 85 ℃, 0 ℃, -50 ℃.***

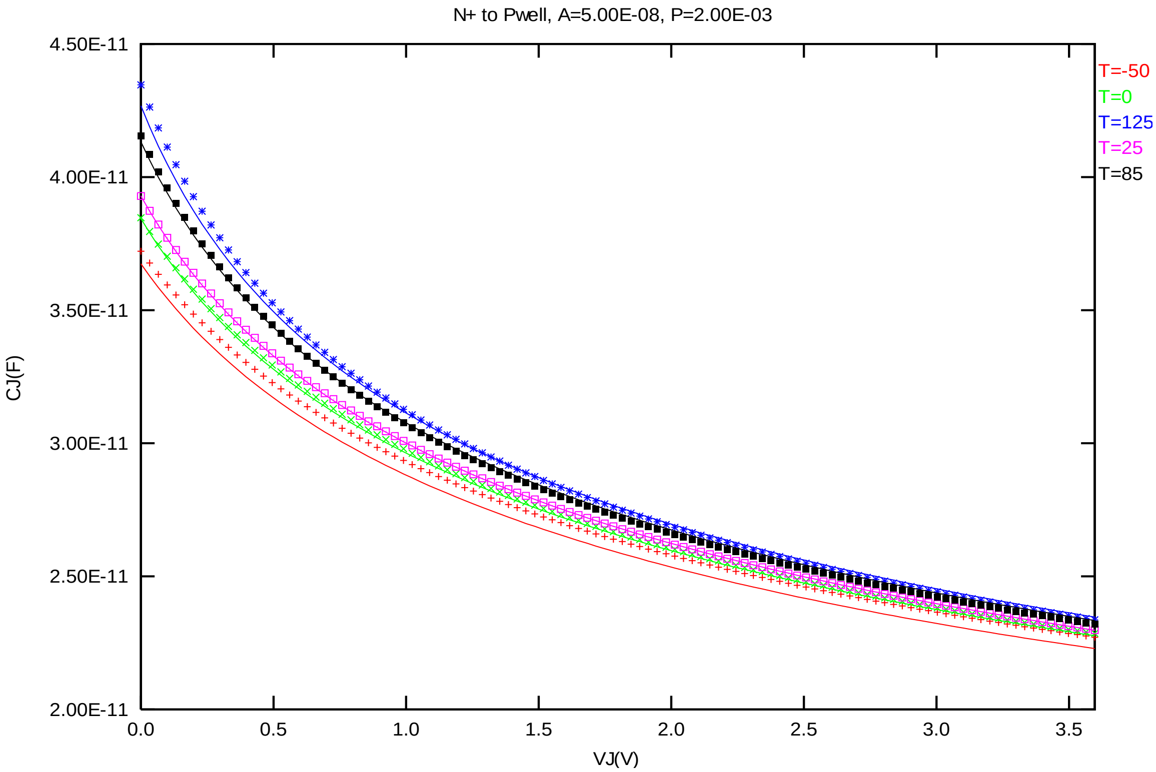


Figure 2-12 Diode junction CV curve at 25 ℃, 125 ℃, 85 ℃, 0 ℃, -50 ℃.

- ***Forward IV curve at 25 ℃, 125 ℃, -50 ℃.***

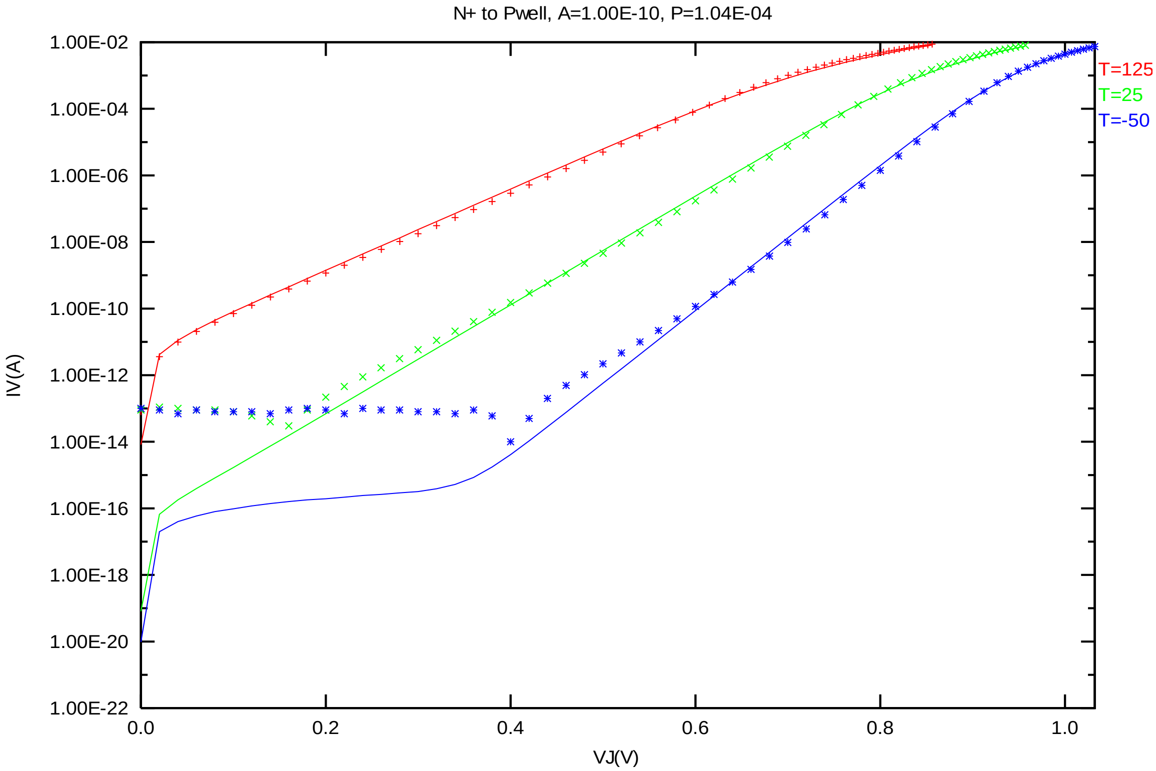


Figure 2-13 Forward IV curve at 25 ℃, 125 ℃ , -50 ℃.

- ***Reverse IV curve at 25 ℃, 125 ℃ ,85 ℃, 0 ℃, -50 ℃.***

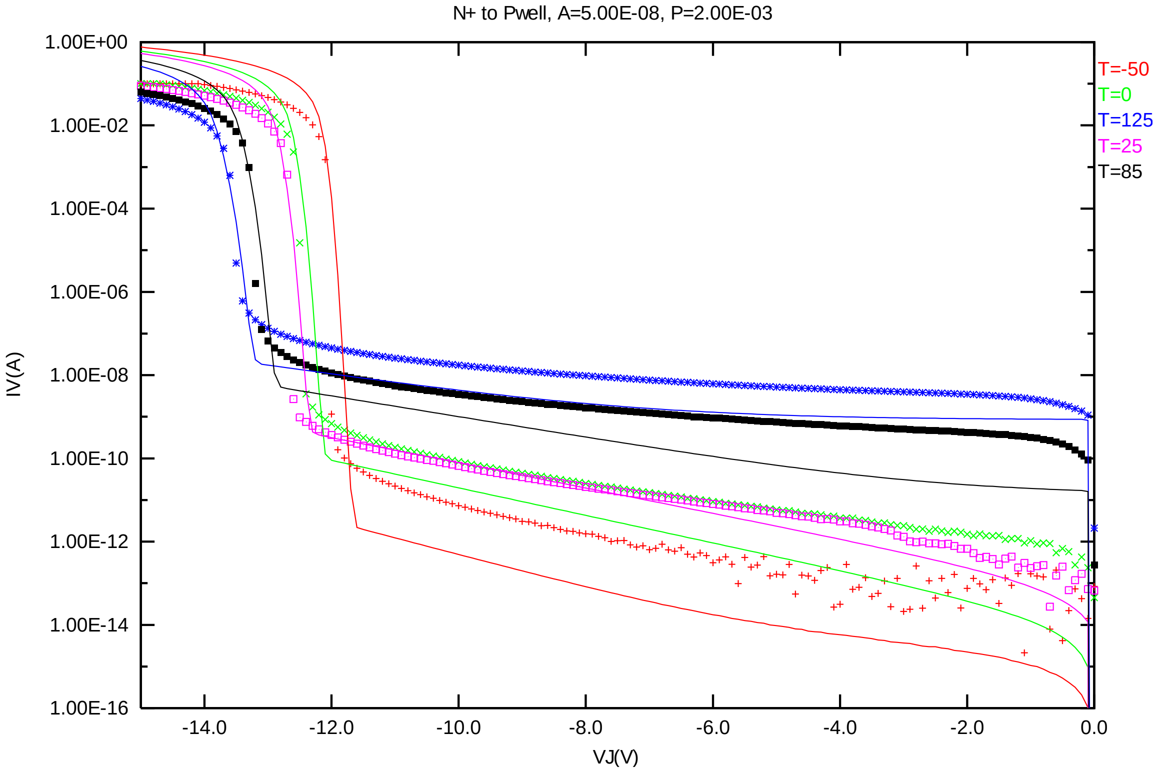


Figure 2-14 Reverse IV curve at 25 ℃, 125 ℃ ,85 ℃, 0 ℃, -50 ℃.

3 2.5V DIOP Diode Model Verification

3.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for DIOP

	DIOP diode model
Mask ID	A11007
Lot ID	D5FHR.11
Wafer Number	#11

This model is characterized as the following geometry and temperature ranges depicted. Device behavior beyond these ranges may not be well described by the model. This model is extracted by devices with these two dimensions :

Junction Area Device :

Area = 5.0e-8 m^2

Perimeter = 2.0e-3 m

Junction Edge Device :

Area = 3.6e-8 m^2

Perimeter = 3.64e-2 m

3.2 Testing Conditions

The model is verified to measurement data under following bias conditions

Forward and reverse diode IV curve:

Table 3-2 Forward and reverse diode IV measurement conditions

Forward Diode

	Start	Stop	Step
V_J(V)	0	1	0.05

Reverse Diode

	Start	Stop	Step
V_J(V)	0	-15	-0.1

Diode junction CV:

Table 3-3 Diode junction CV measurement conditions

V_{BIAS} [V]	VH: nwell & VL: p+ **From 0 to 3.63 V in 0.033 V step**
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50, 0, 25, 85, 125

3.3 2.5V DIOP model verification

3.3.1 2.5V DIOP model verification at room temperature (25 ℃)

- ***Diode junction CV curve at 25 ℃.***

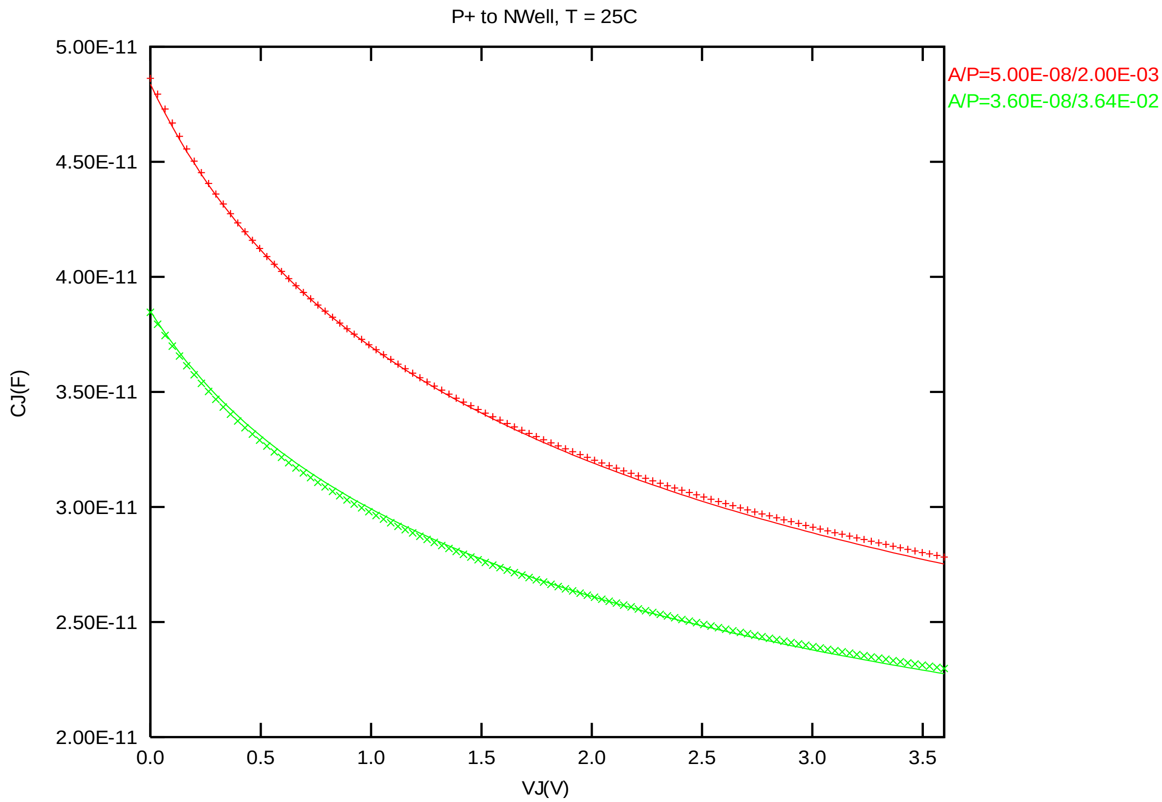
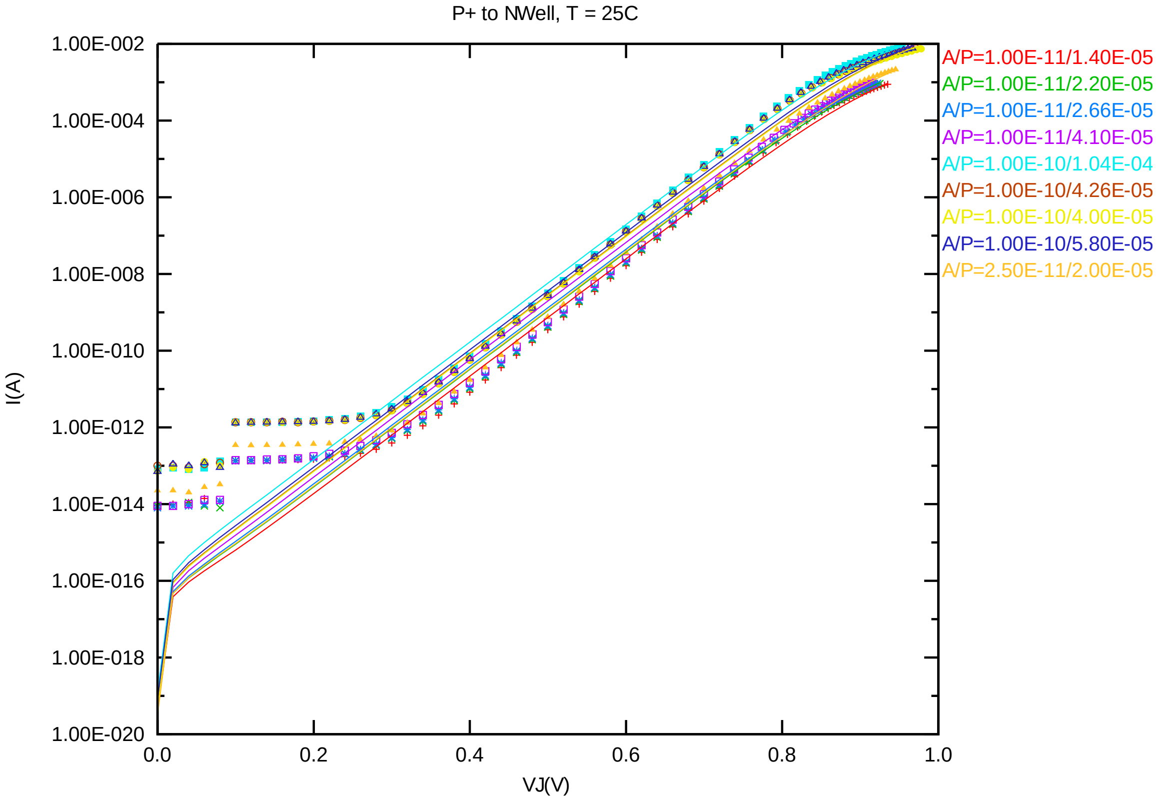


Figure 3-1 Diode junction CV curve at 25 ℃

- ***Forward IV curve at 25 ℃.***



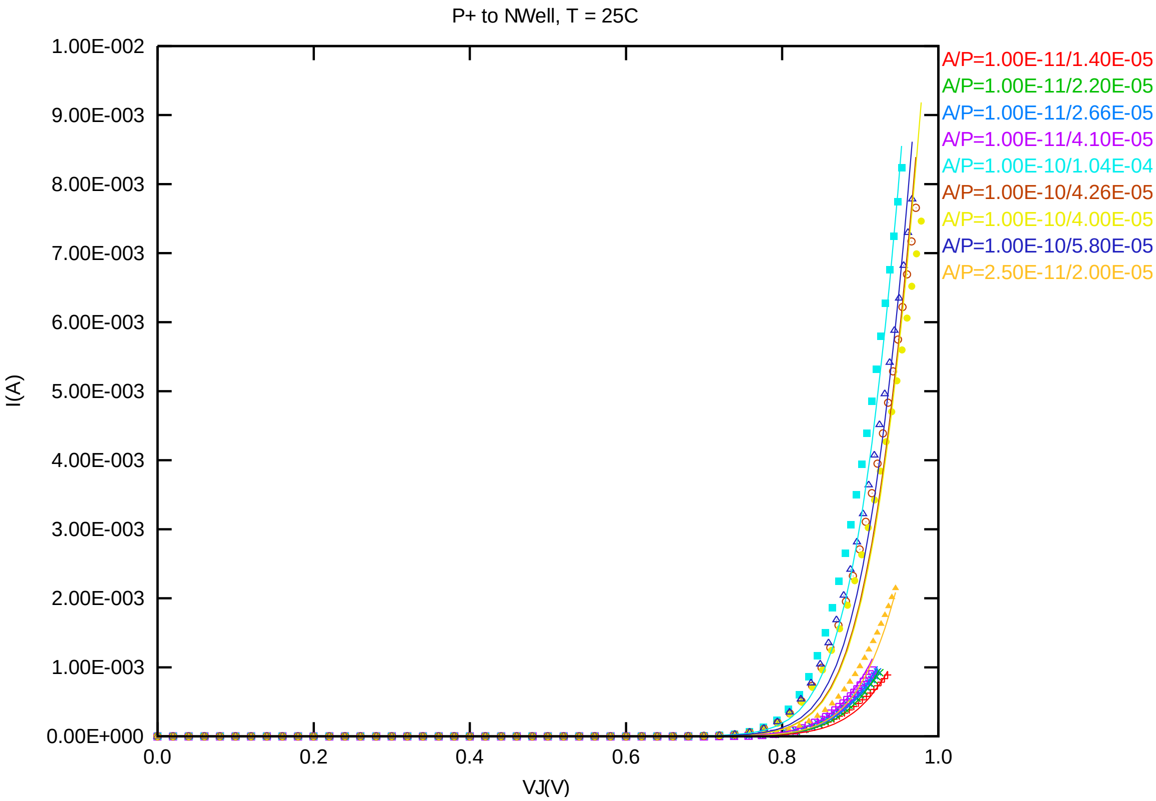


Figure 3-2 Forward IV curve at 25 ℃

- ***Reverse IV curve at 25 ℃.***

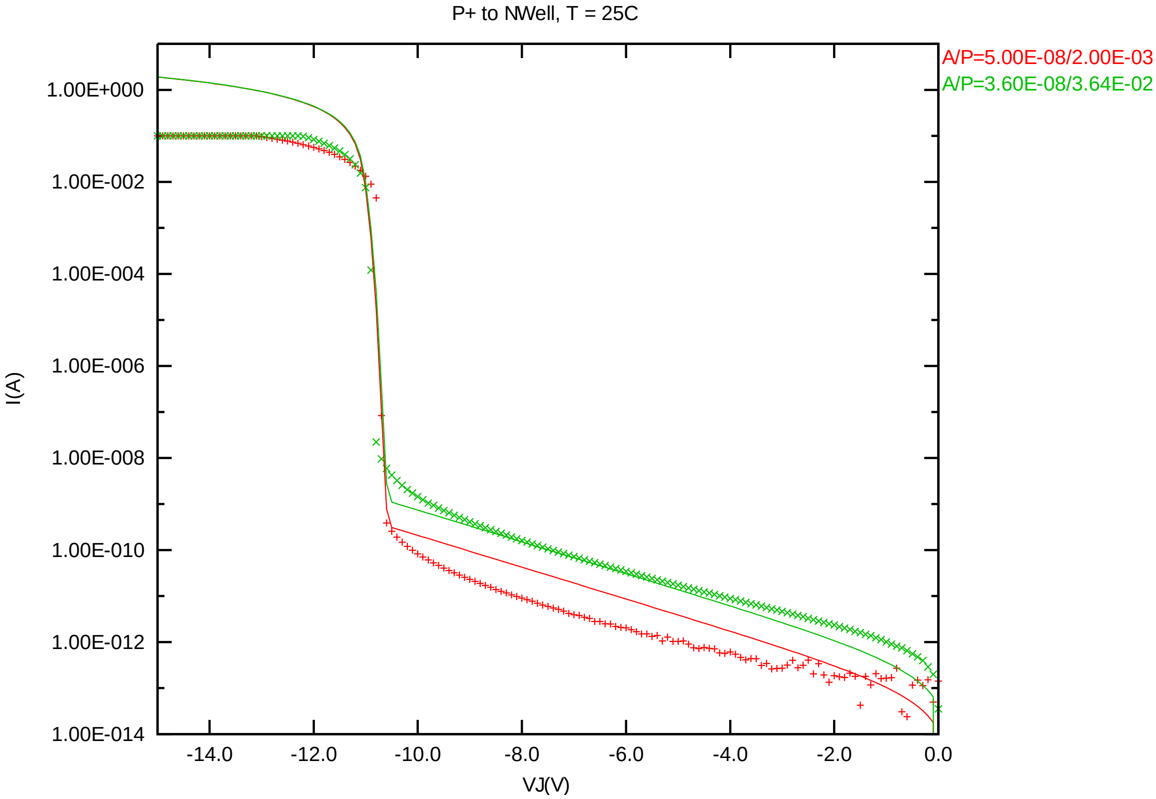


Figure 3-3 Reverse IV curve at 25 ℃

3.3.2 2.5V DIOP model verification at 125 ℃

➢ ***Diode junction CV curve at 125 ℃.***

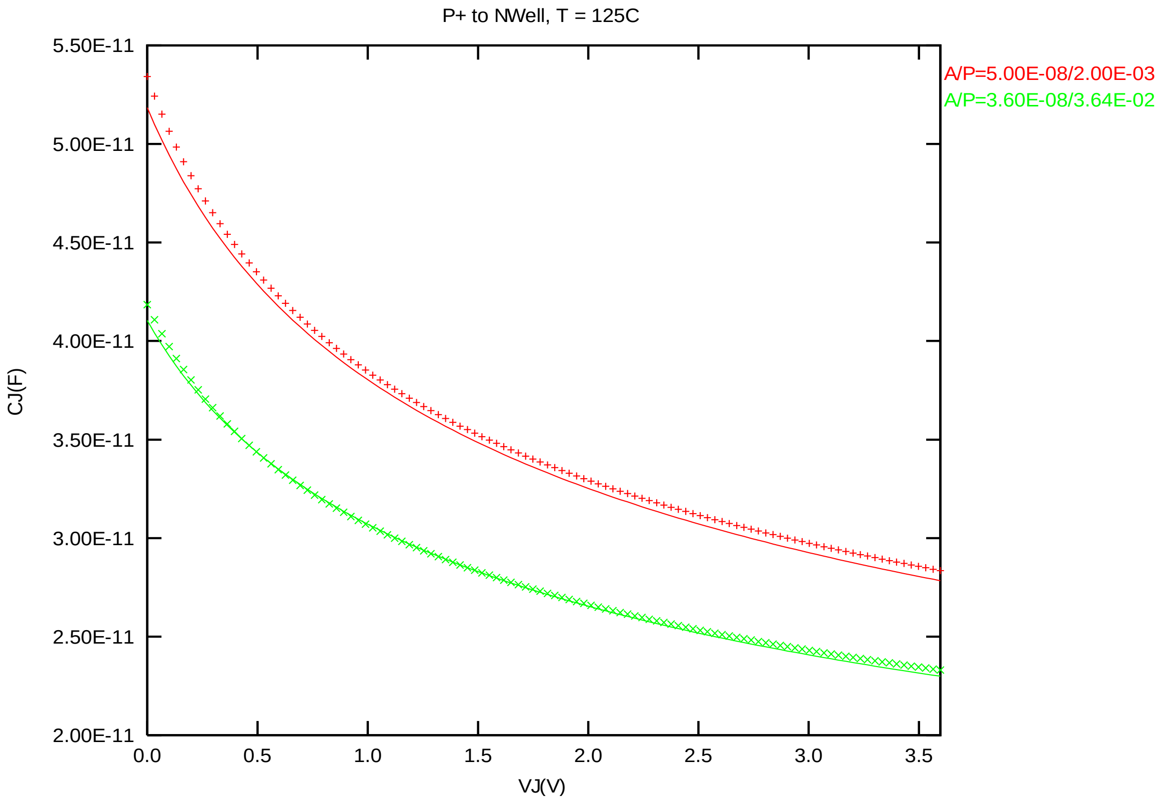
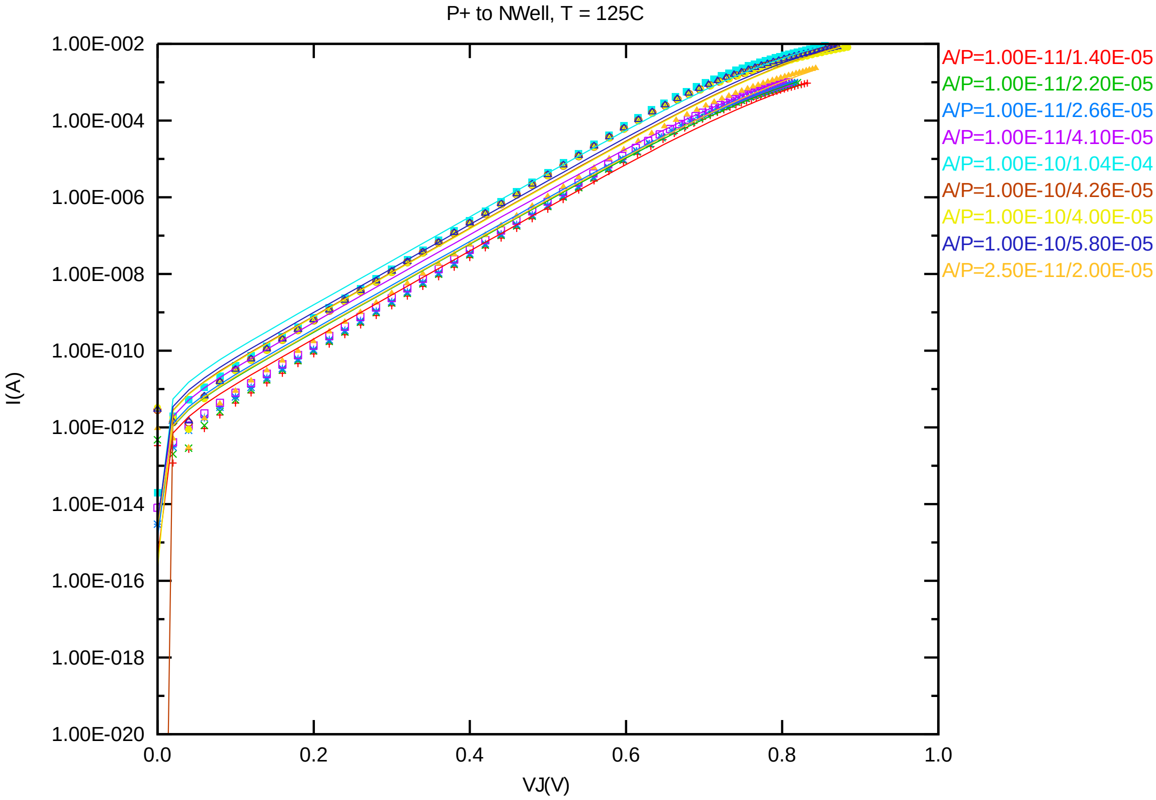


Figure 3-4 Diode junction CV curve at 125 ℃

➢ ***Forward IV curve at 125 ℃.***



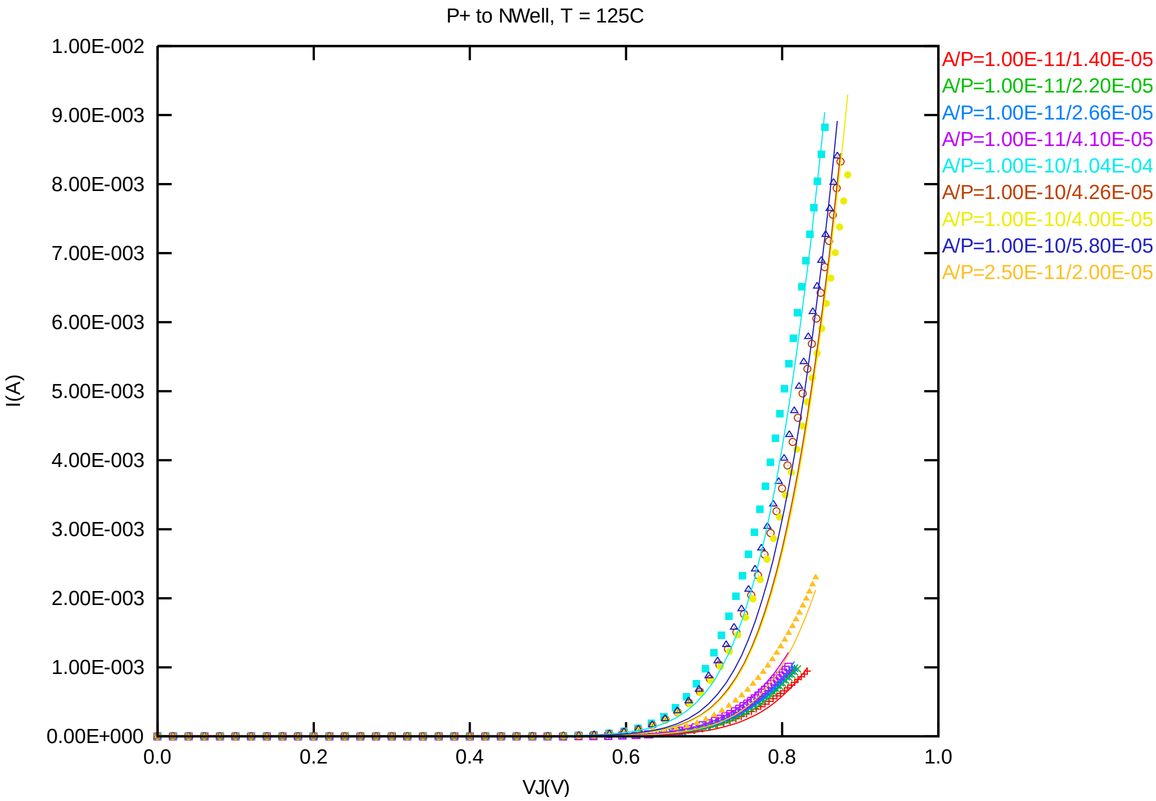


Figure 3-5 Forward IV curve at 125 ℃

- ***Reverse IV curve at 125 ℃.***

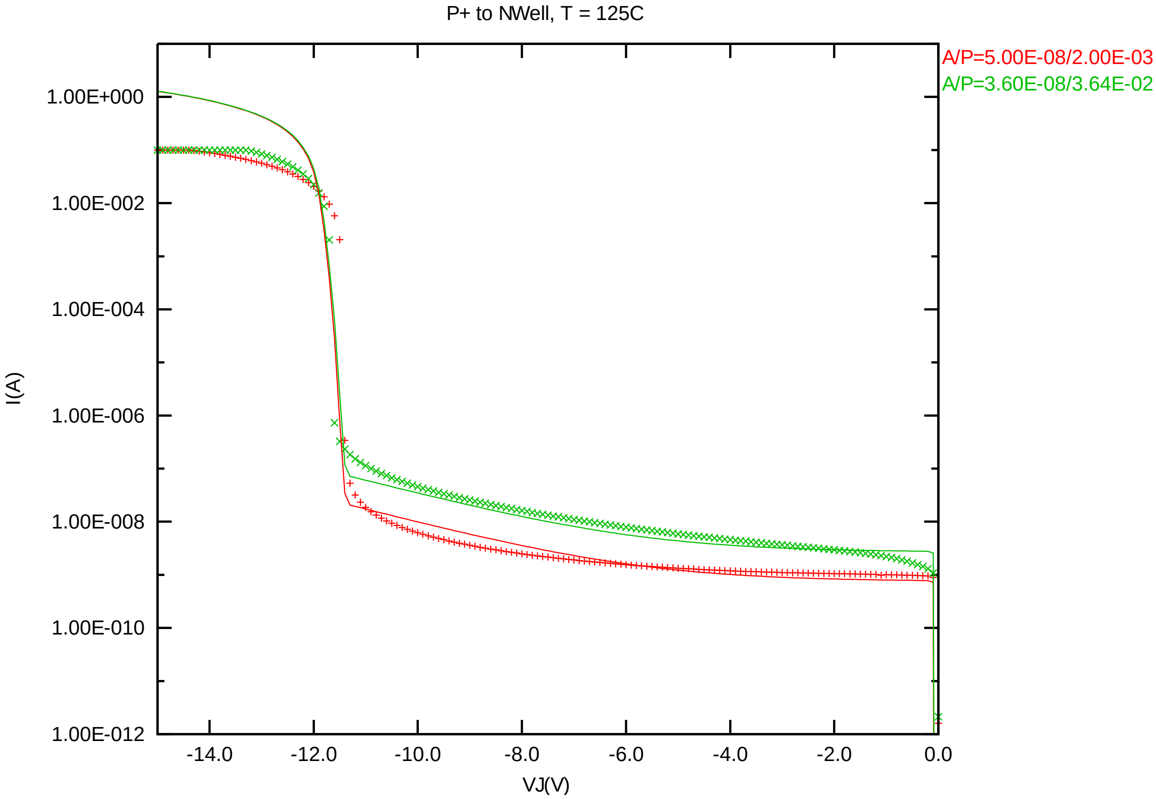


Figure 3-6 Reverse IV curve at 125 ℃

3.3.3 2.5V DIOP model verification at 85 ℃

➢ ***Diode junction CV curve at 85 ℃.***

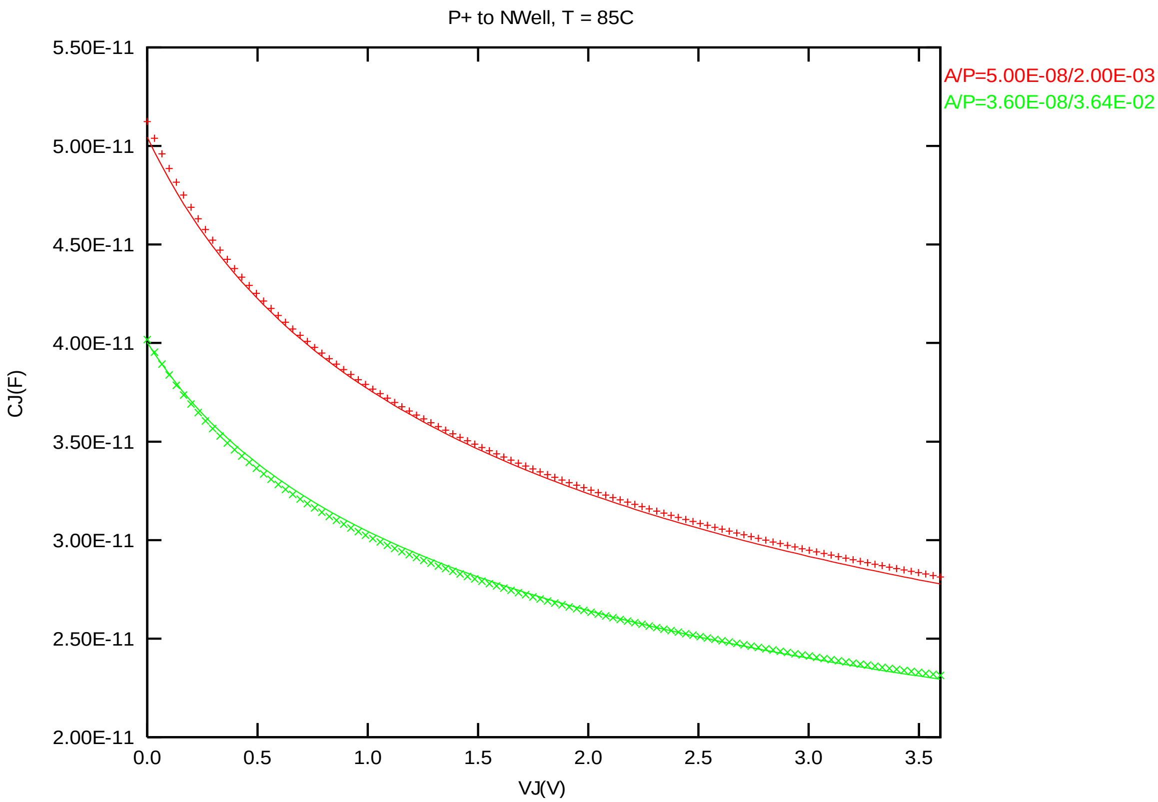


Figure 3-7 Diode junction CV curve at 85 ℃

3.3.4 2.5V DIOP model verification at 0 ℃

➢ ***Diode junction CV curve at 0 ℃.***

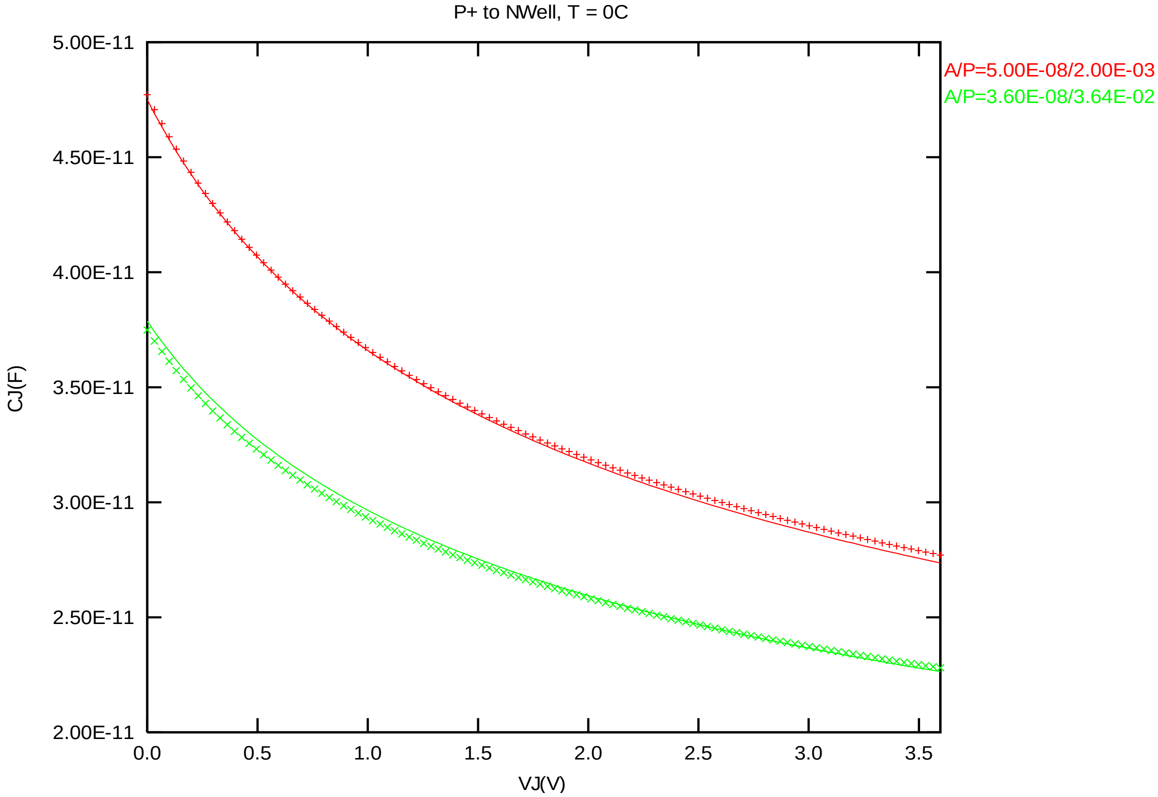


Figure 3-8 Diode junction CV curve at 0 ℃

3.3.5 2.5V DIOP model verification at -50 ℃

➢ ***Diode junction CV curve at -50 ℃.***

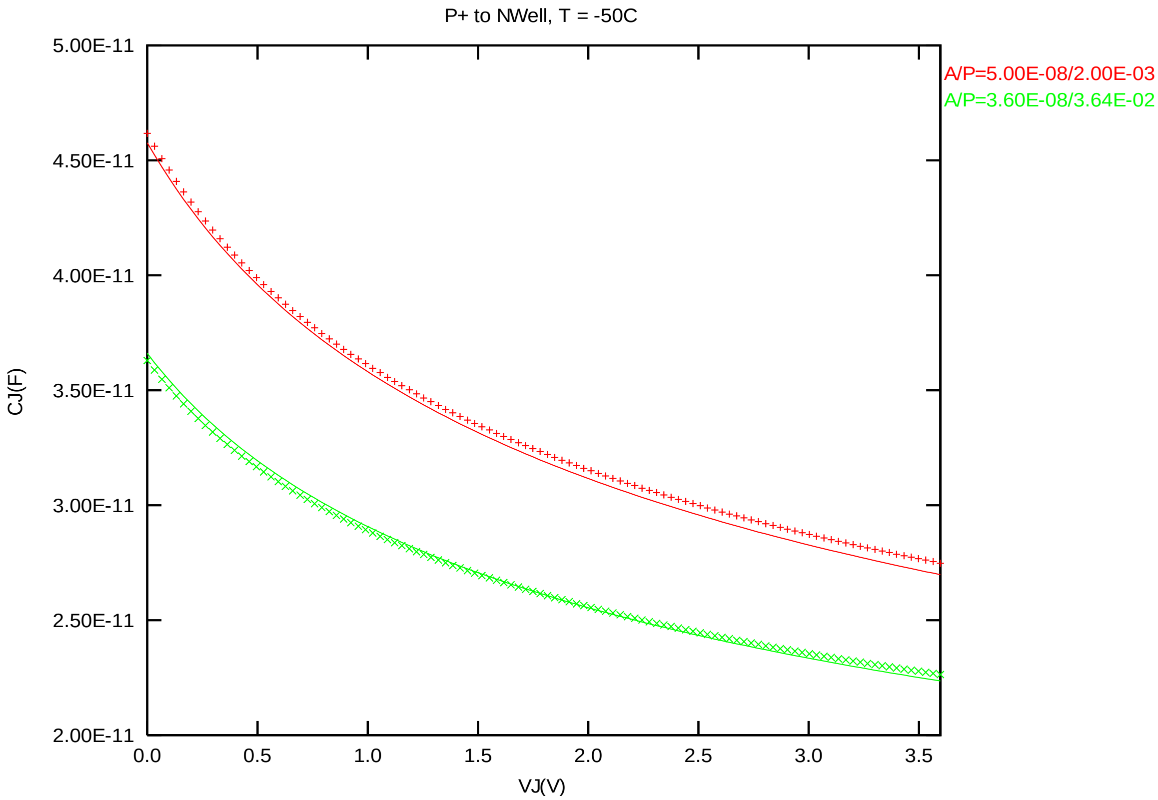
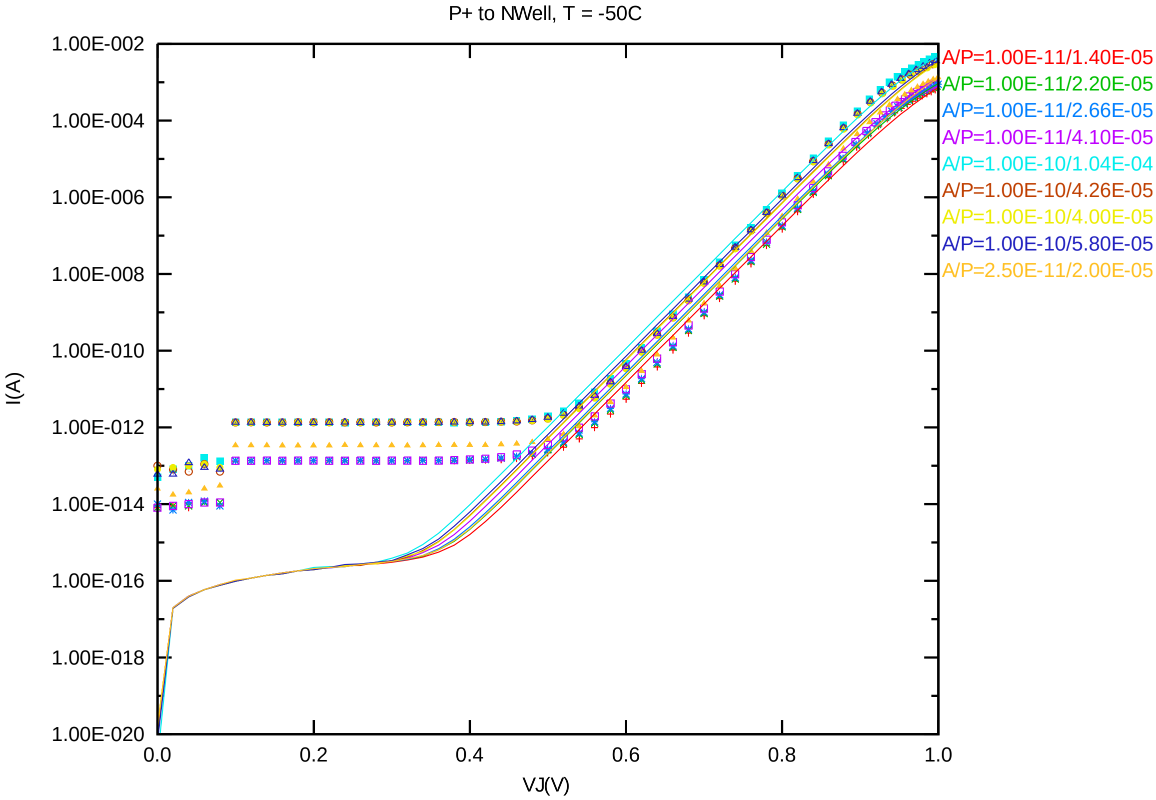


Figure 3-9 Diode junction CV curve at -50 ℃

- ***Forward IV curve at -50 ℃.***



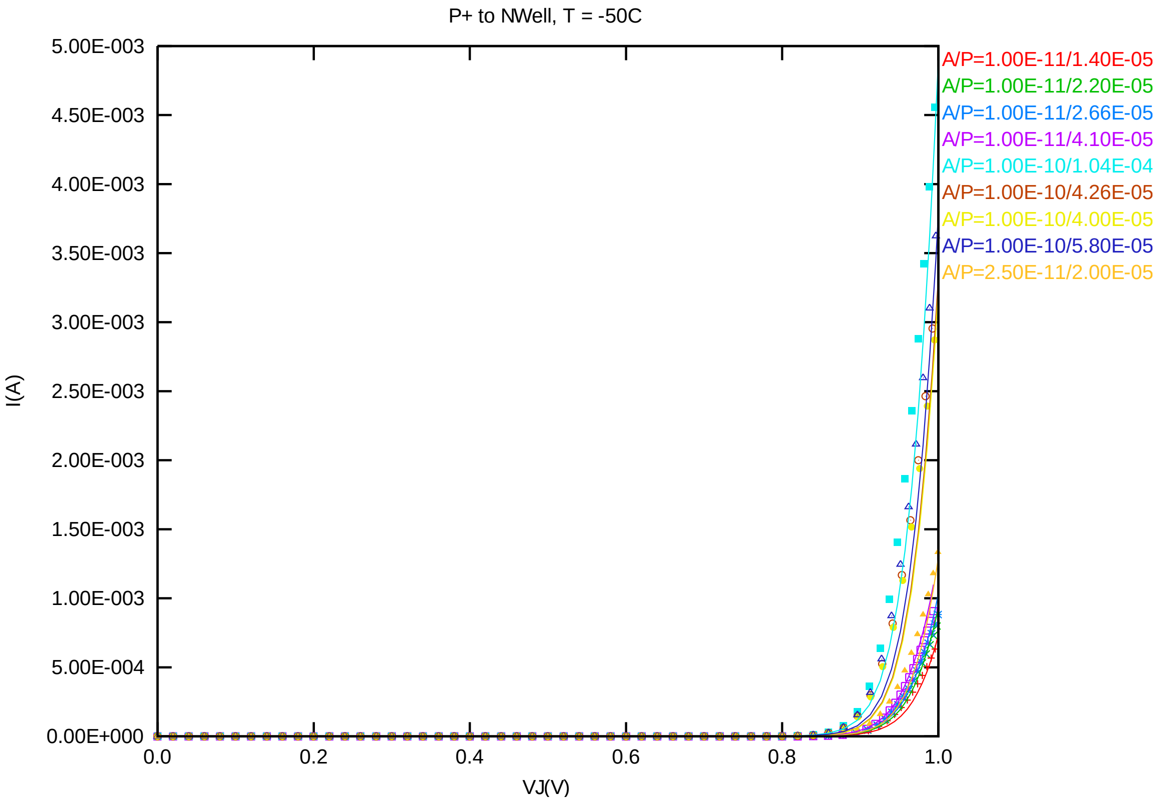


Figure 3-10 Forward IV curve at -50 ℃

- ***Reverse IV curve at -50 ℃.***

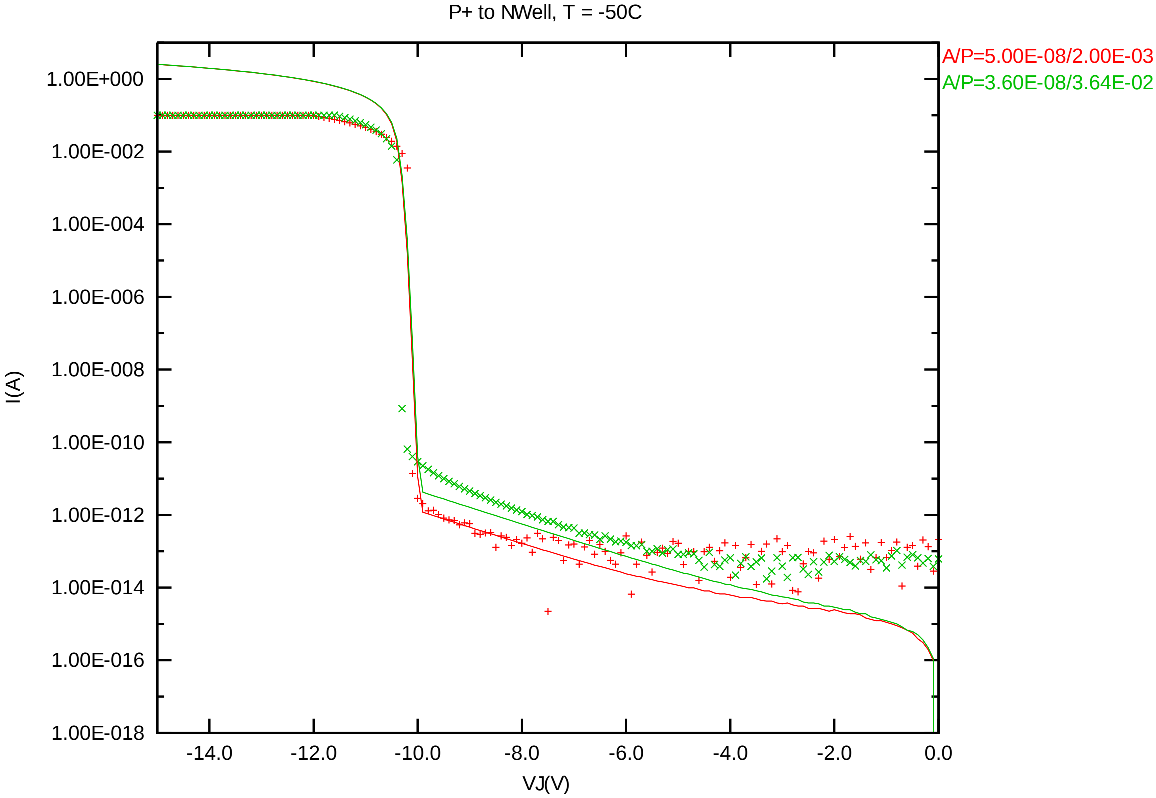


Figure 3-11 Reverse IV curve at -50 ℃

3.3.6 2.5V DIOP model verification for temperature effect

- ***Diode junction CV curve at 25 ℃, 125 ℃, 85 ℃, 0 ℃ -50 ℃.***

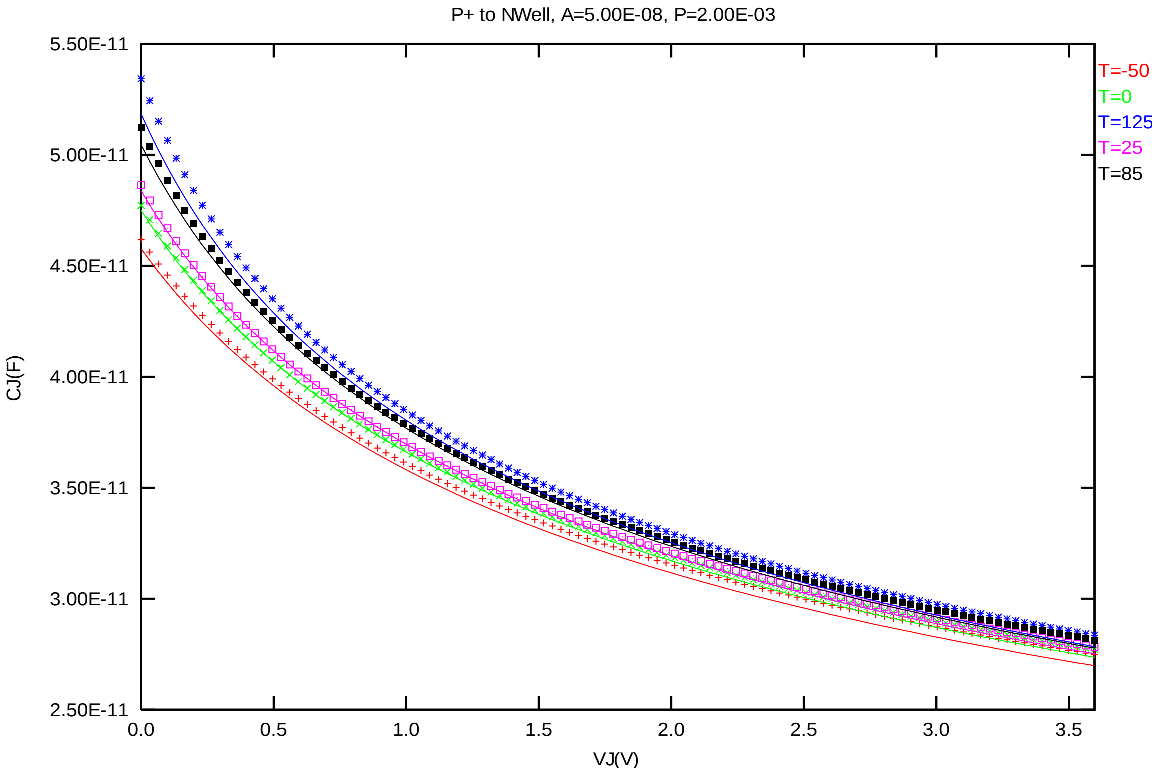


Figure 3-12 Diode junction CV curve at 25 ℃, 125 ℃, 85 ℃, 0 ℃ -50 ℃.

➢ ***Forward IV curve at 25 ℃, 125 ℃, -50 ℃.***

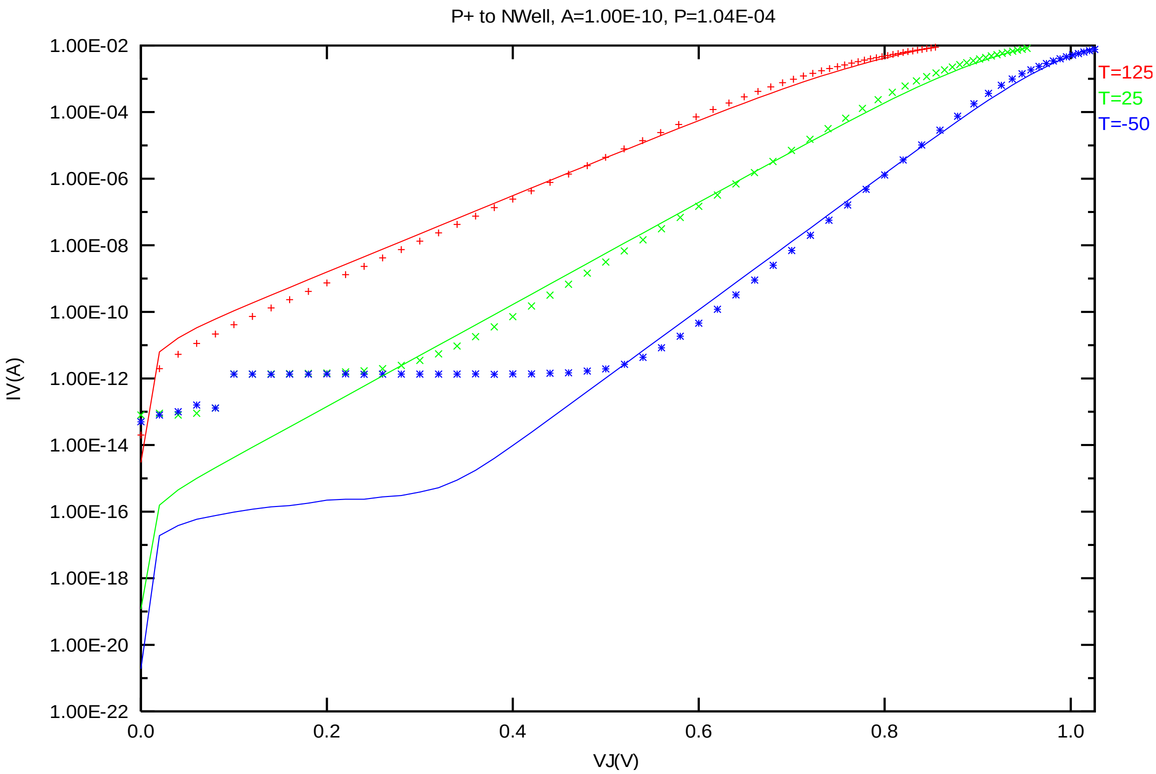


Figure 3-13 Forward IV curve at 25 ℃, 125 ℃, -50 ℃

- ***Reverse IV curve at 25 ℃, 125 ℃, 85 ℃, 0 ℃ -50 ℃***

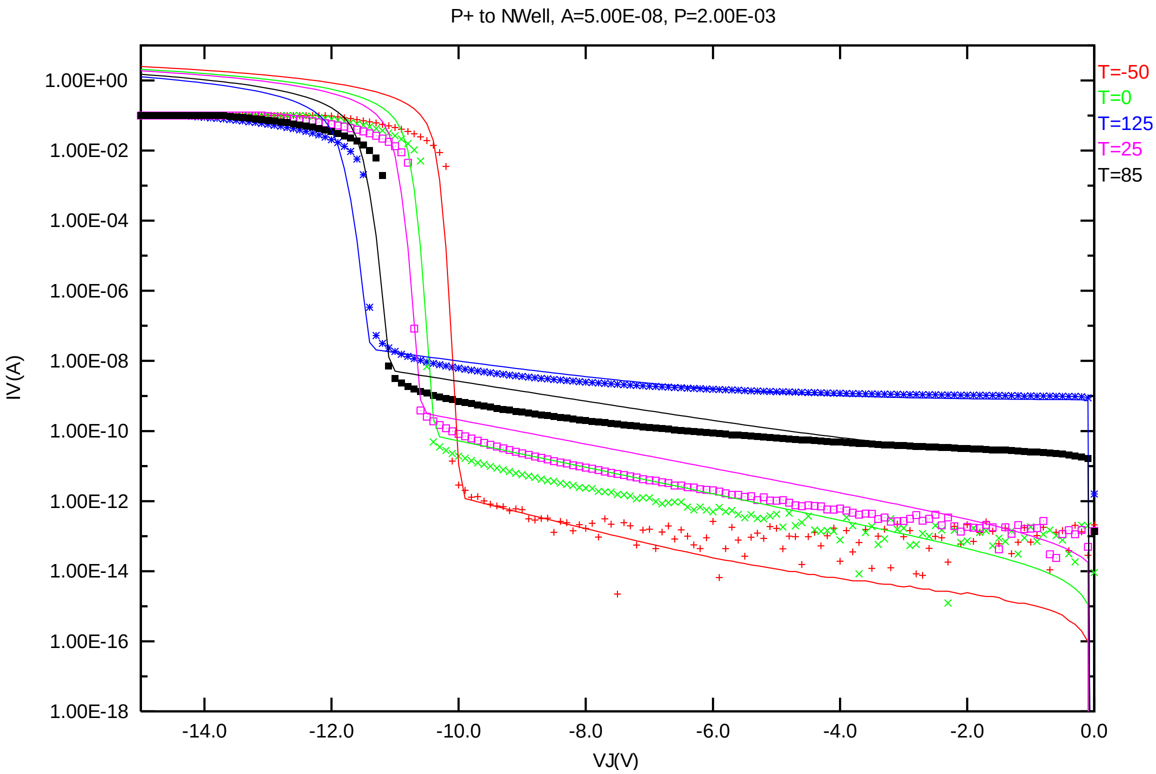


Figure 3-14 Reverse IV curve at 25 ℃, 125 ℃, 85 ℃, 0 ℃ -50 ℃

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

4.2 ELDO Warning Message

Table 4-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

4.3 SPECTRE Warning Message

Table 4-3 Spectre warning message

NO.	Warning Message	Explanation
1.	None	None